BITS Pilani – M.Tech (AIML) – Machine Learning – Assignment 2

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1. GitHub Repository Link

https://github.com/2026aj05790-apeksha/ml-assignment2

2. Live Streamlit App Link

https://ml-assignment2-bxkybtx2fackxbtb4bh9vk.streamlit.app

3. Screenshot – assignment execution on BITS Virtual Lab

[Insert ONE BITS Virtual Lab screenshot here]
Set 'screenshot' in config.json and rebuild.

4. README Content

■ Breast-Cancer Diagnosis – Five-Classifier Comparison & Streamlit Deployment

BITS Pilani WILP · M.Tech (AIML) · Machine Learning · Assignment 2

a. Problem statement

Early, accurate discrimination between **malignant** and **benign** breast tumours from fine-needle-aspirate (FNA) cell-nucleus measurements reduces unnecessary biopsies and prevents missed cancers.

The task is a **binary classification** problem: build, evaluate and compare five classifiers (Logistic Regression, Decision Tree, KNN, Naive Bayes, SVM) on one dataset using six metrics

(Accuracy, AUC, Precision, Recall, F1, MCC), then deploy an interactive Streamlit app that evaluates the *saved* models on an uploaded test CSV without retraining.

b. Dataset description

Item	Detail
Source	Breast Cancer Wisconsin (Diagnostic) – UCI Machine Learning Repository (Wolberg, Street & Mangasarian, 1995), loaded via `sklearn.datasets.load_breast_cancer`
Instances	569 (train 455 / test 114, stratified 80:20, seed 42)
Features	30 numeric (>= 12 required): 10 nucleus measurements × {mean, standard error, worst}
Target	`diagnosis` – 0 = malignant (212, 37.3 %), 1 = benign (357, 62.7 %)
Class balance	Moderately imbalanced (1 : 1.68) → MCC and recall reported alongside accuracy
Missing values	None
Pre-processing	`StandardScaler` inside each sklearn `Pipeline` (fit on training data only – no leakage); the saved `.joblib` therefore contains the complete preprocessing

The 10 base measurements: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension.

Many are strongly collinear (radius–perimeter–area r > 0.95), which matters for Naive Bayes (see observations).

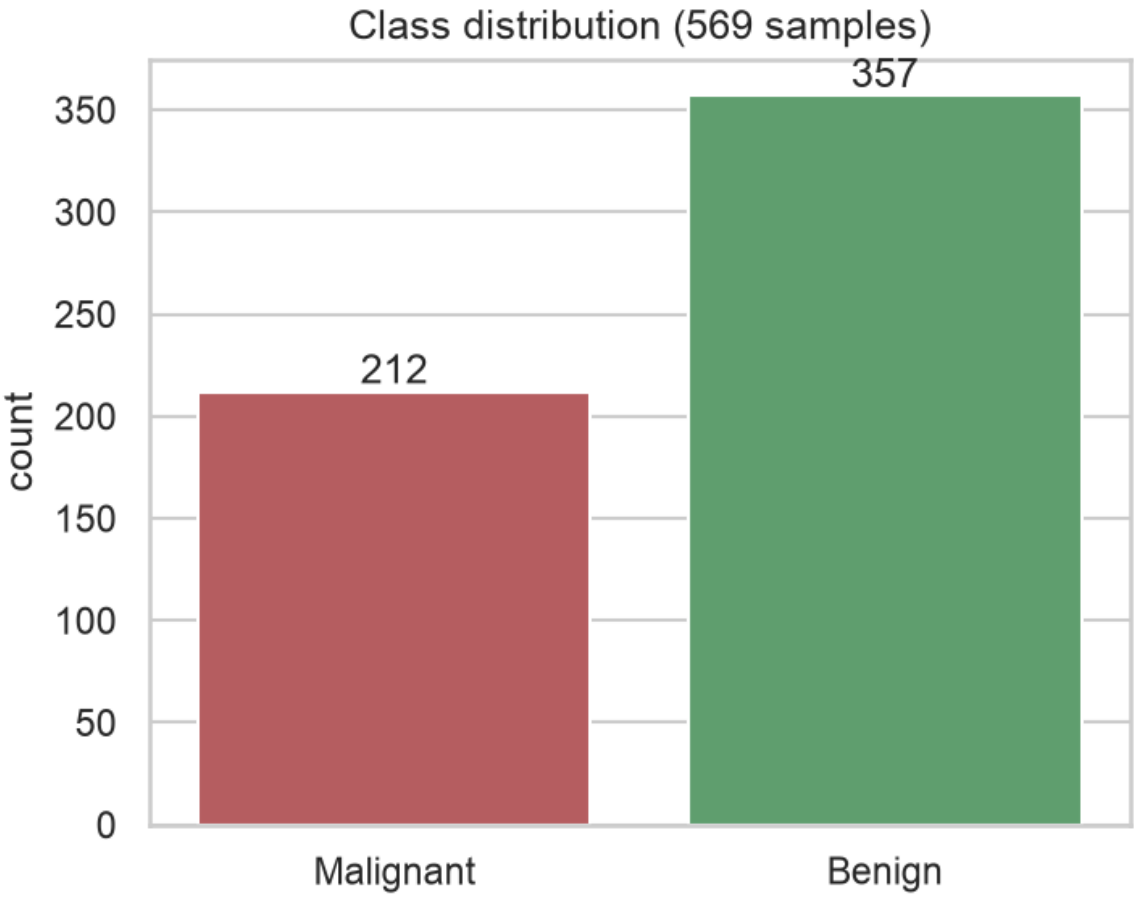


Figure 1 – Class distribution

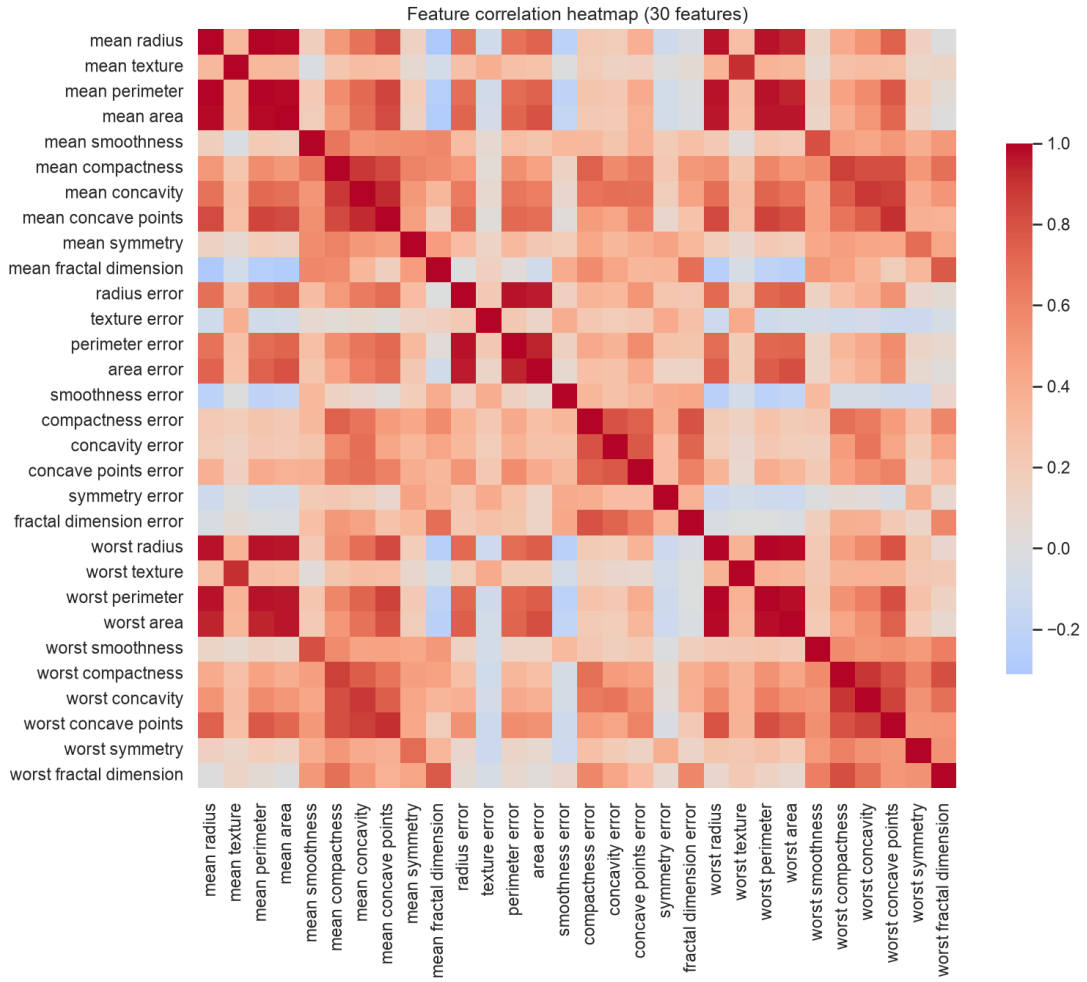


Figure 2 – Feature correlation heatmap: blocks of highly correlated size features

c. GitHub Repository Link

https://github.com/2026aj05790-apeksha/ml-assignment2

```
project-folder/
|-- app.py                  # Streamlit app (upload CSV → choose model → metrics, confusion matrix, report, comparison)
|-- train.py                # trains 5 models, computes metrics, saves models + figures
|-- build_readme.py         # regenerates this README from real results
|-- requirements.txt
|-- test_data.csv           # 114-row hold-out test set (30 features + diagnosis)
|-- full_dataset.csv        # complete 569-row dataset (for reference)
|-- model/                  # 5 saved pipelines (.joblib)
|-- results/                # metrics.csv, confusion_matrices.json
|-- figures/                # all plots used below
`-- agents/                 # grader agent + improvement loop (development tooling)
```

Live Streamlit app: https://ml-assignment2-bxkybtx2fackxbtb4bh9vk.streamlit.app

How to run / reproduce

```
pip install -r requirements.txt
python train.py             # retrains, rewrites model/, results/, figures/, test_data.csv
streamlit run app.py        # launches the evaluation app
```

d. Models used

All five models are trained on the **same** 455-row training split and evaluated on the **same** 114-row hold-out test set.

Hyper-parameters: LR (C=1), Decision Tree (max_depth=5, min_samples_leaf=3), KNN (k=7, distance-weighted), Gaussian NB (defaults), SVM (RBF, C=1, gamma='scale').

Comparison table (test set)

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9825	0.9954	0.9861	0.9861	0.9861	0.9623
Decision Tree	0.9035	0.9373	0.9420	0.9028	0.9220	0.7969
KNN	0.9737	0.9884	0.9600	1.0000	0.9796	0.9442
Naive Bayes	0.9298	0.9868	0.9444	0.9444	0.9444	0.8492
Support Vector Machine	0.9825	0.9950	0.9861	0.9861	0.9861	0.9623

Generalisation check (5-fold stratified CV on training data vs train vs test)

ML Model Name	Train Acc	5-fold CV Acc	Test Acc
Logistic Regression	0.9890	0.978 ± 0.010	0.9825
Decision Tree	0.9802	0.916 ± 0.019	0.9035
KNN	1.0000	0.963 ± 0.020	0.9737
Naive Bayes	0.9385	0.934 ± 0.029	0.9298
Support Vector Machine	0.9824	0.969 ± 0.015	0.9825

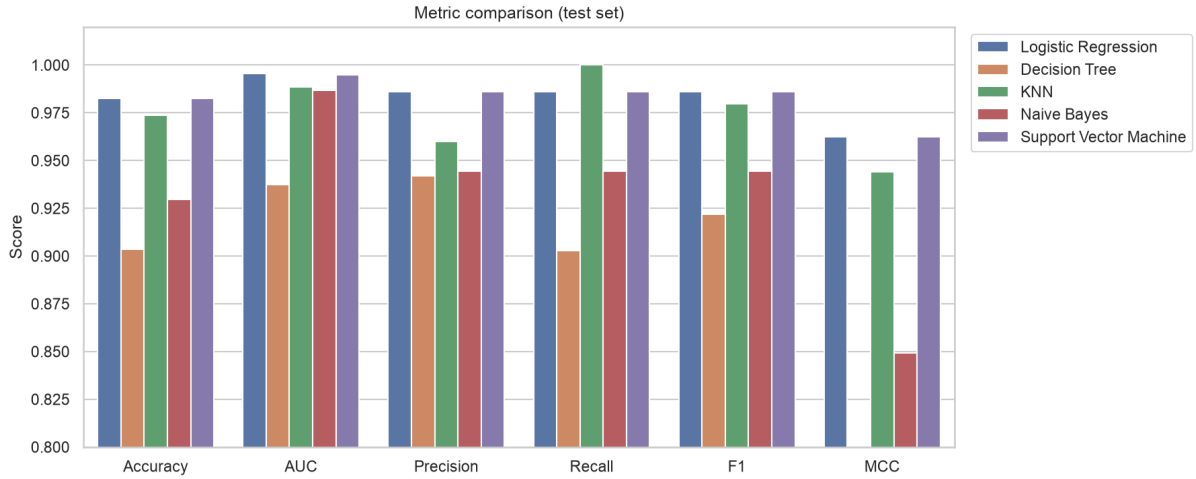


Figure 3 – Metric comparison across models

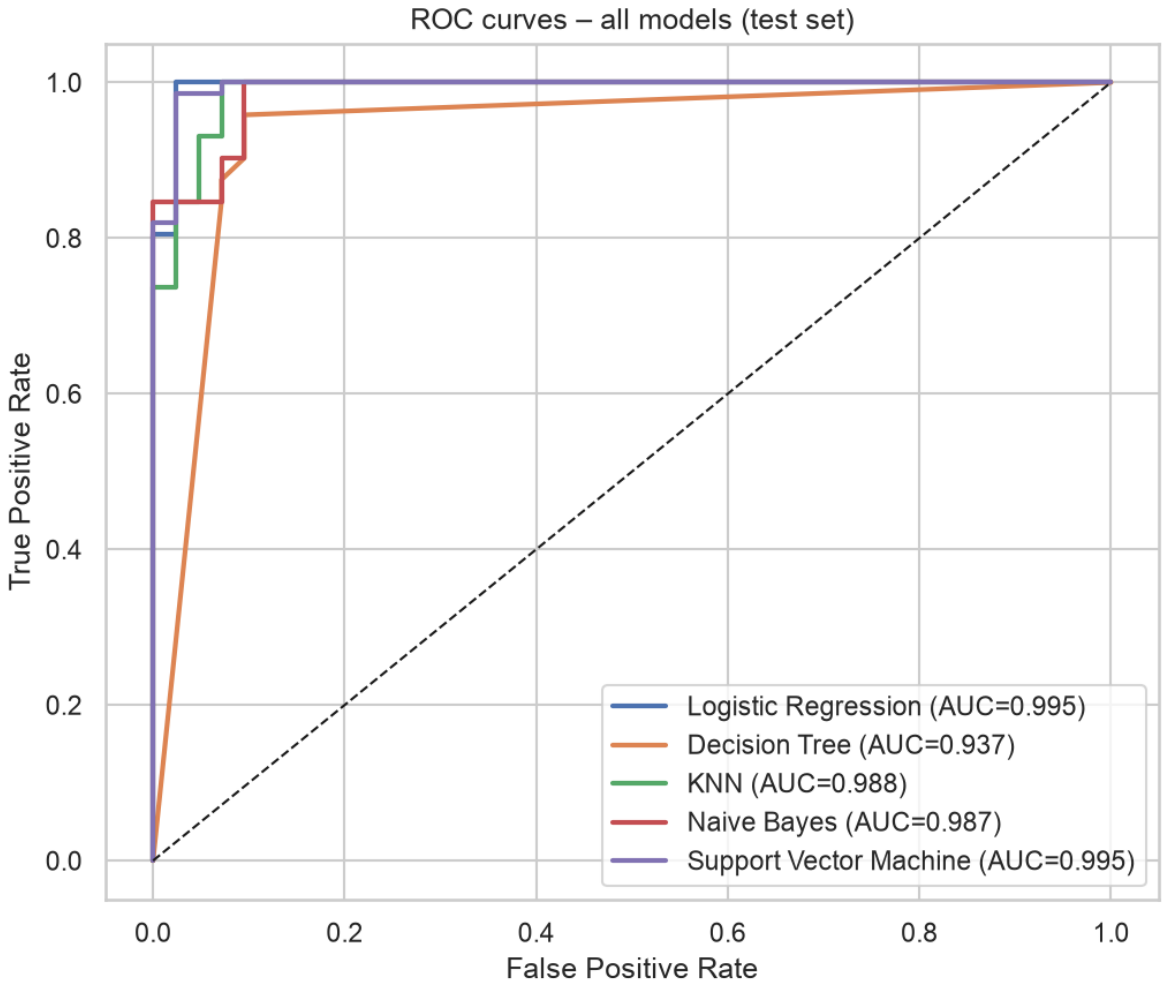


Figure 4 – ROC curves (test set)

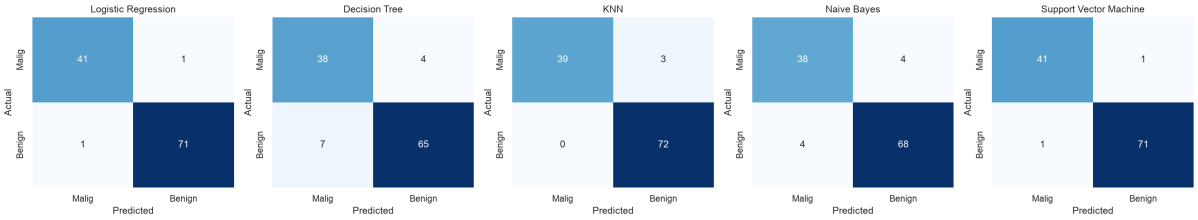


Figure 5 – Confusion matrices (class 0 = malignant)

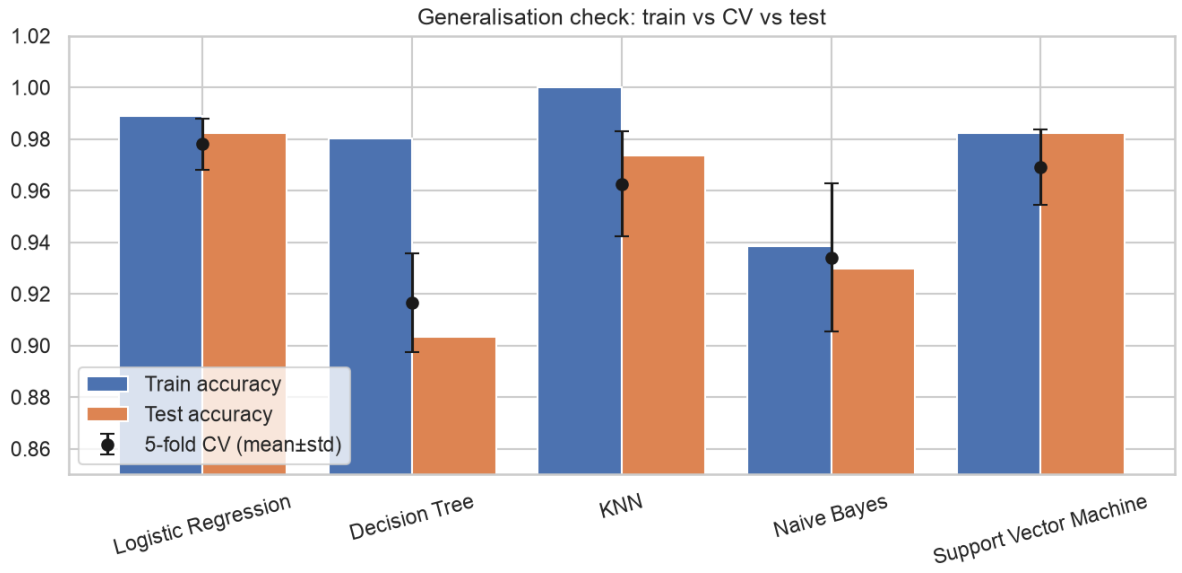


Figure 6 – Train vs CV vs test accuracy

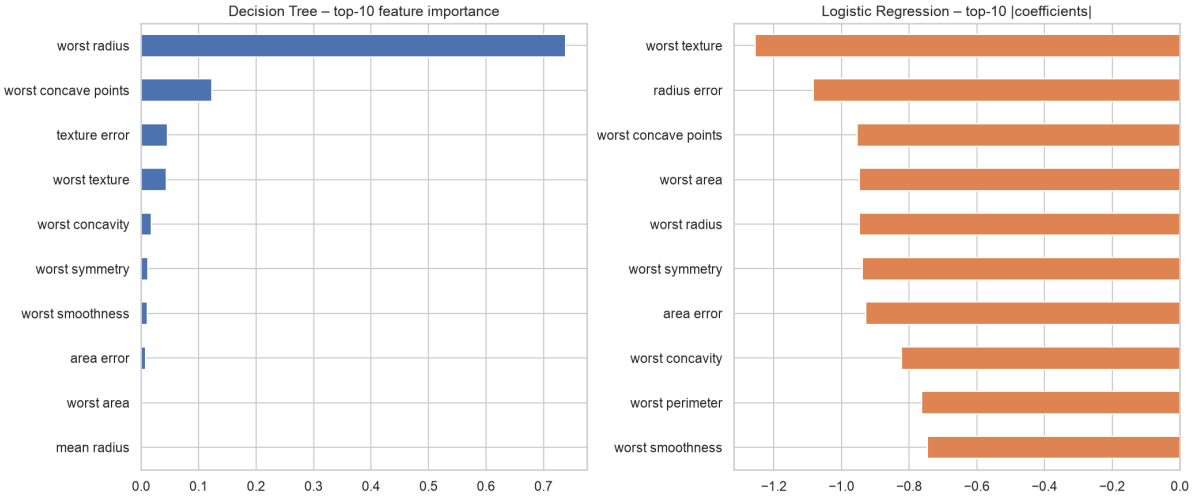


Figure 7 – Feature importance (Decision Tree) and coefficients (Logistic Regression)

Observations on model performance

ML Model Name	Observation about model performance
Logistic Regression	Reaches 98.25% accuracy, AUC 0.9954 and MCC 0.962 (tied best), with only 2 errors out of 114 test rows. This works because the malignant/benign classes are almost linearly separable in standardised feature space, so a linear boundary has low bias and, with only 30 inputs, low variance. Train accuracy (0.989) is close to CV (0.978 ± 0.010), indicating no overfitting. Coefficients are directly interpretable (see feature-importance figure).
Decision Tree	Weakest model: accuracy 90.35%, AUC 0.937, MCC 0.797. Train accuracy (0.980) far exceeds test accuracy, a classic high-variance overfitting signature even with max_depth=5 and min_samples_leaf=3, because axis-aligned splits must approximate a diagonal boundary with many small steps and 455 training rows are too few to stabilise them. It missed 4 malignant cases (recall 0.903), clinically the worst outcome. Its coarse probability estimates also pull AUC down. Benefit: fully transparent rules.
KNN	Accuracy 97.37% and perfect recall (1.000) with 3 missed malignant cases and 0 false positives, so precision (0.960) is the cost. Distance-weighted k=7 on standardised features works because similar tumours cluster in feature space; without scaling it would be dominated by area/perimeter features. Train accuracy is 1.000 (distance weighting memorises training points) while CV is 0.963 ± 0.020, so the CV standard deviation is the largest of the distance-based models and prediction is slower since it stores all data.

ML Model Name	Observation about model performance
Naive Bayes	Accuracy 92.98%, MCC 0.849 but a strong AUC of 0.9868, meaning it ranks cases well yet its 0.5 cut-off is poorly calibrated. The independence assumption is violated here because radius, perimeter and area (and their 'worst' versions) are correlated above 0.95 (see heatmap); correlated evidence is counted multiple times, making posteriors overconfident. Train (0.939) and test accuracies are similar, so it suffers from bias rather than variance. It is the fastest model and needs no tuning.
Support Vector Machine	Ties with Logistic Regression on every test metric (accuracy 98.25%, AUC 0.9950, MCC 0.962). The RBF kernel maximises the margin and handles any mild non-linearity, and scaling makes gamma='scale' meaningful. CV accuracy (0.969 ± 0.015) is slightly below logistic regression, suggesting the extra flexibility buys nothing on this nearly linear problem while costing training time and interpretability.
Overall, Winner for your dataset?	**Logistic Regression** (with SVM as a statistical tie). LR and SVM have identical test accuracy (98.25%), MCC (0.962) and recall, but LR has the higher 5-fold CV accuracy (0.978 ± 0.010 vs 0.969 ± 0.015), the highest AUC, trains in milliseconds and is interpretable. We rank by MCC and recall rather than accuracy alone, because MCC is robust to the 63/37 class imbalance and a false negative (missed malignancy) is the costliest error. KNN has the best recall but lower precision/MCC. With only 114 test rows, one sample moves accuracy by 0.88%, so LR vs SVM differences are not significant.

Error analysis

15 of 114 test samples are misclassified by at least one model and **1** by all five – these are genuinely borderline tumours whose measurements overlap the other class.

Most of the accuracy gap between models therefore comes from a small set of ambiguous cases, which is why MCC/recall and CV are used in addition to a single test split.

Streamlit application features

CSV upload (test data only) · model dropdown · six metric cards · confusion matrix + classification report · ROC curve · adjustable decision threshold ·

side-by-side comparison of all five saved models on the uploaded data · prediction download · input validation (missing columns / non-binary target).

Models are loaded with `joblib`; **no training happens in the app**.

Limitations

• Single public dataset with 569 rows; a 114-row test set gives wide confidence intervals (one sample ≈ 0.88 % accuracy), so LR vs SVM differences are not statistically significant.

• Hyper-parameters were set by reasoned defaults and light CV, not an exhaustive search; data come from one institution, so external generalisation is unverified.

• Models are decision-support tools, not diagnostic devices.

Future work

• Nested cross-validation with grid/Bayesian search, probability calibration (Platt/isotonic) – especially for Naive Bayes.

• Ensembles (Random Forest, Gradient Boosting, stacking), SHAP explanations, external validation on another cohort.

• Cost-sensitive threshold selection to explicitly minimise false negatives.